



# An unfished area enhances a spiny lobster, *Panulirus argus*, fishery: implications for management and conservation within a Biosphere Reserve in the Mexican Caribbean

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**Abstract** The Caribbean spiny lobster, *Panulirus argus*, Latreille, is the main source of income for the communities in the Sian Ka'an Biosphere Reserve Mexico. The fishery has recently been certified as sustainable by the Marine Stewardship Council provided that further stock assessment is carried out. A total of 379 lobsters were tagged in an unfished area offshore from the Bahía del Espíritu Santo fishing grounds to assess whether lobsters remained within these areas and were thus fully protected. The lobsters recaptured in the shallow area (5.3%) were sufficient to develop a multistate mark recapture model, which takes into account fishing and natural mortality, tag reporting rate and tag loss. This estimated that between 15 and 20% of all adult lobsters dwelling in the unfished area moved into the fishery and were subjected to exploitation. This study suggests that the offshore unfished area provides protection to the majority of the stock in this area while adding to and maintaining fishing yields within the inshore commercial fishery.

**KEY WORDS :** community based conservation, coral reef, fisheries, population dynamics.

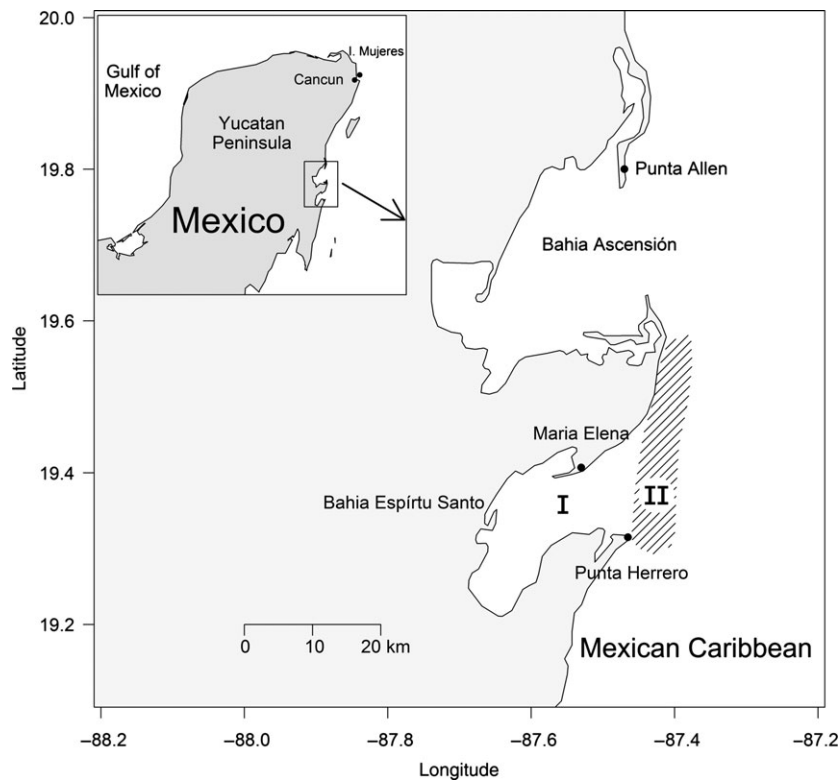
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## Introduction

Marine protected areas (MPAs) are often defined as no take zones where fishing is prohibited. As part of the conservation efforts to sustain exploited species such as lobsters, MPAs are designed to preserve important habitats that serve as shelter, foraging grounds or movement

corridors, as well as protecting the breeding stocks and increasing the fishery yield of the target species (Acosta 1999; Goñi *et al.* 2010). In Biosphere Reserves like Sian Ka'an (SK-BR see map rectangular insert Fig. 1), where fishing is allowed but access is restricted and operations only occur within areas of less than 20 m depth (MRAG-Americas 2012; Ley-Cooper *et al.* 2013),

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**Figure 1.** Map showing Mexico and the Bahía del Espíritu Santo study area, which is the southern bay of the Sian Ka'an Biosphere Reserve (the reserve area is identified by the rectangle on the map insert). Bahía de la Ascensión is neighbouring to north and the Caribbean sea to the east of the Yucatan Peninsula. Areas are marked as 'I' the fished shallow bay (<20 m deep) and 'II' the unfished offshore area (>20 m deep) respectively.

management initiatives are also expected to replenish stocks by increasing biomass and abundance. However, few empirical studies have demonstrated whether these management initiatives optimise population viability, and how they may affect benthic dispersal dynamics of the Caribbean spiny lobster *Panulirus argus*, Latreille (Acosta 1999).

Assigning no take areas where fishing is totally banned is being promoted as a conservation strategy for heavily exploited species such as *P. argus*, although there is generally only limited evidence that movements of adult individuals from unfished areas replenish the populations in areas subject to fishing (spillover effect) (Russ *et al.* 2004; Goñi *et al.* 2010). This is also the case in the SK-BR where unfished offshore areas deeper than 20 m are unofficially conceived as no take areas of the MPA. These areas are especially valuable for sustainability as they are predominantly populated by mature-sized lobsters (Lozano-Álvarez *et al.* 1993; Ley-Cooper *et al.* 2013). Understanding both recruitment and lobster movement patterns from the deep unfished area into the fishery can help evaluate the Biosphere Reserve's potential to meet management performance standards, by providing information about habitat use, home range, migrations,

retention times/spillover and location of spawning grounds (Goñi *et al.* 2006; Bertelsen 2013). Given the complexity of the post-larval settlement and recruitment patterns of *P. argus* (Briones-Fourzán *et al.* 2008), the only tangible fishery benefit that can be demonstrated in favour of the deep unfished areas is the recruitment of adults which replenish the shallow fishing grounds (<20 m) and therefore enhance fishing yields (Russ & Alcalá 1996; Bevacqua *et al.* 2010). This is usually examined using tagging experiments (Frusher & Hoenig 2003; Goñi *et al.* 2010; Bertelsen 2013).

Spiny lobster is the most valued single species fished in the Caribbean and the main economic source for families living in the Sian Ka'an Biosphere Reserve, Mexico (SK-BR). The marine segment of the SK-BR is an MPA, where fishing permits for lobsters is restricted to three cooperatives with exclusive concession rights within both existing bays – Bahía de la Ascensión and North and Bahía Espíritu Santo-South (Fig. 1) (Ley-Cooper *et al.* 2013). Fishing operations are confined to the shallow areas (<20 m) west from the barrier reef due to restrictions in fishing gear and regulations, as only free diving is allowed and the use of any sort of alternate air source or deep trap is banned (Sosa-Cordero

*et al.* 1999; Ley-Cooper *et al.* 2013). As a result, only the portion of the whole lobster stock lying within shallow waters is exploited (area I, Fig. 1), and the deep offshore area (>20 m) is effectively a no take zone which is unfished (area II, Fig. 1). This limitation is a main criterion for the sustainable management of this fishery. However, the MSC certification process identified the multiple sources of recruitment variation and movement of adult lobsters to and from the deep areas as one of the main sources of uncertainty in assessing the stock status (MRAG-Americas 2012).

*Panulirus argus* displays movement behaviours throughout its life cycle for many purposes, including relocating to more appropriate habitat and foraging for food and reproduction (Acosta 1999; Briones-Fourzán *et al.* 2003; Ríos-Lara *et al.* 2007). After settlement within shallow vegetated areas, juveniles attain a transitional size (typically >15–20 mm carapace length CL) at which they begin to seek shelters (rock crevices, holes and ledges; undercut coral heads and sponges). Emigration out to deeper reef areas for reproductive purposes begins to occur in pre-adult lobsters (size ~75 mm CL).

How the physical characteristics of the deep segment (>20 m) of the SK-BR MPA interacts to support a viable population of *P. argus* in the shallows has been the focus of much discussion, but few empirical studies have investigated the mechanisms which affect adult dispersal, distribution or abundance – information important for designing protected areas (Acosta 1999; Bertelsen 2013). Little is known about the population dynamics of lobsters that dwell in the deep platform and the offshore areas ( $\geq 20$  m) of the SK-BR (Lozano-Álvarez *et al.* 1993; Ley-Cooper *et al.* 2013). It is possible that some adult lobsters from the deep water stock may migrate into the shallow areas as a result of dispersal movements, for reasons such as returning after spawning (Bevacqua *et al.* 2010; Bertelsen 2013); seasonal behavioural changes (Herrnkind 1980; García *et al.* 1991; González-Cano 1991; Lozano-Álvarez *et al.* 1993); or in search of food and shelter (Ríos-Lara *et al.* 2007). To date, the proportion of lobsters that might move inshore from deep areas has not been estimated, and the assumption that the adult stock is fully protected in deeper waters should be re-examined for this BR.

More robust stock assessment and the establishment of biological reference points have been suggested as means to improve the management and evaluation of the lobster fisheries in SK-BR, and now form a condition for the on-going sustainability certification granted by the Marine Stewardship Council (MSC) (MRAG-Americas 2012). In this fishery, exploitation rates are high within the shallow bays (depths < 20 m, area I), and it has been suggested that, to sustain such catch rates,

lobsters recruitment to this fishery must be frequently replenished by growth of juveniles and/or by an input of lobsters moving in from unfished areas such as those found offshore in deeper waters (>20 m) (González-Cano 1991; Lozano-Álvarez *et al.* 1991a,b, 1993; Briones-Fourzán *et al.* 2007; Ley-Cooper *et al.* 2013).

This study addresses the question of how the present management regulations, which constrain fishing to depths less than 20 m, may affect benthic dispersal dynamics of the Caribbean spiny lobster and their catch rates within the Sian Ka'an Biosphere Reserve. A multi-state mark recapture model was used to test the hypothesis that an autumn season migration takes place, with a proportion of lobsters moving from the offshore (>20 m) unfished areas of the MPA into the shallow areas which are fished. The impact this has on fishing yields is examined and discussed.

## Materials and methods

### Study area

The study was conducted in the Mexican Caribbean, in the central segment of the coast of Quintana Roo/Yucatan Peninsula within the SK-BR MPA. Here, the continental shelf is narrow, not exceeding 4 km from the coast and ending at depths averaging 50–60 m, after which depths rapidly reach >400 m (Lozano-Álvarez *et al.* 1991a,b). The SK-BR encompasses two large marine embayment's named Bahía del Espíritu Santo (south) and Bahía de la Ascensión (north) (Fig. 1). Bahía del Espíritu Santo is a large and shallow bay ranging in depths from 1–20 m, with an area of approximately 300 km<sup>2</sup> (Sosa-Cordero *et al.* 1999). Within this bay, there is a limited access fishery exclusively exploited by two cooperatives and comanaged by environmental and fishery federal government authorities (Sosa-Cordero *et al.* 2008).

For the purposes of this study, the bay was divided into two areas (Fig. 1): area I, the commercially exploited shallow bay (<20 m) to the west and area II, the unexploited offshore deeper area (>20 m) to the east. In practice, area II is a no take zone within this BR, as diving with the use of alternative air sources (e.g. SCUBA) is prohibited, although lobsters (*P. argus*) are naturally distributed in both areas (Wahle & Fogarty 2006).

### Description of the fishery and local stock

The lobster fishery is based on concessions to two cooperatives which fish using a 'Casita/Campo' system. Casitas are large artificial shelters that can harbour the full size range of lobsters and are allotted to individual fisher

families associated with the cooperative. The bay and fishing areas are divided into parcels called Campos and are hence owned by the family who manage them in a semi-ownership arrangement (Lozano-Álvarez *et al.* 1991a,b; Briones-Fourzán *et al.* 2000; Sosa-Cordero *et al.* 2008). Permits are renewed annually, and cooperatives must comply with the federal fishing regulations, which include a closed season from March 1 to June 30, a minimum tail size of 13.5 cm (~74.5 mm, CL; Lozano-Álvarez *et al.* 1991a,b) and prohibition of capturing egg-bearing females.

Lobsters are caught mainly in the bays within the fringing reefs using small boats and by skin diving to a maximum depth of 20 m (Ley-Cooper *et al.* 2013). For the purpose of this study, the local stock is referred to as the proportion of the population found within the bay, including the adjacent water areas found beyond the reef and to the east of Bahía del Espíritu Santo (depths  $\leq 100$  m) (Briones-Fourzán & Lozano-Álvarez 2001; Briones-Fourzán *et al.* 2007).

#### Lobster tagging and recapture

A total of 379 lobsters (56 in August and 323 in September, 243 females and 136 males) were tagged and released where they were caught, in water depths which ranged between  $>20$  and  $<40$  m, in area II (Fig. 1), during the first 5 days of each month in 2011. All lobsters were caught using hand nets or snares while fished using SCUBA, and were tagged with T-Bar tags (Hallprint, Australia) in their ventral abdomen region as described by Ley-Cooper *et al.* (2013). The size range of tagged lobsters was 70.1–140.5 mm CL. Fishing location, sex and size (CL) were recorded for each lobster.

Recaptures were obtained from both licensed fishing boats and the cooperative depots during the fishing seasons 2011/2012 and 2012/2013. Research observers went on-board boats for 10 days during every month of the fishing season. Additionally, information on tag reporting in log books was obtained at reception points.

#### Multistate tag-recapture model

A multistate tag-recapture model (MSTR) based on a model framework described by Hilborn (1990) was developed to estimate the monthly movement rates of tagged lobsters from an unfished area (area II) to a fished area (area I) during each month of the fishing season following the tag releases (i.e. August 2011–January 2012). The model, built in 'R' (R Development Core Team 2012), encompassed only the first fishing season following release (2011/12) as this period covered the majority of tag returns. Returns in the 2012/2013 season

were relatively few, and they provided little information for the MSTR model to determine movement rates with any degree of confidence.

A model was custom-built for this study because previously developed models, such as those found in packages like 'MARK' (White & Burnham 1999), required tagging and recaptures to occur in both areas in its multistate design. In this study, lobsters were only tagged and released in the unfished deep area  $>20$  m (area II) and recaptured only in the shallow fished area  $<20$  m (area I). The MSTR model incorporated fishing and natural mortality, reporting rate and tag loss. It differed slightly from that described by Hilborn (1990), which also included natural mortality, and all mortality was treated as instantaneous, and a four-dimensional array rather than a three-dimensional array was used to track lobsters.

The model consisted of four main components, which combined allowed the likelihood of recapturing a tagged lobster in the shallow fished area to be determined. These components represented the processes of tag-release, migration, mortality (fishing/natural/tag loss) and likelihood of recoveries. The model employed a monthly time-step (August–January) and considered the numbers of tagged lobsters at liberty and recaptured in each time interval starting in the calendar month when lobsters were first released (August 2011) and terminating at the calendar month when the last recaptured lobster of that fishing season was recorded (January 2012). The model grouped lobsters into two areas: (I) the fished shallow waters ( $<20$  m) and (II) the offshore unfished deeper waters ( $>20$  m). The model contained a series of assumptions:

- Tag loss would occur at a rate previously determined from aquaria studies (Ley-Cooper *et al.* 2013). The exponential decay relationship describing the monthly proportion of tags retained was  $TL = 0.953 * e^{(-0.151 * L)}$ , where TL is the proportion of lobsters still tagged and L is the number of months after release. This rate of tag loss lies within the ranges obtained by other lobster studies (Forcucci *et al.* 1994; Montgomery & Brett 1996; Sharp *et al.* 2000; Dubula *et al.* 2005; Ehrhardt 2008).
- Commercial fishing effort was homogenous across the entire fishery.
- Each lobster within the model had the same probability of being captured.
- During the study, tagged lobsters will not migrate twice, i.e. will not enter the fishery and then leave.
- Natural mortality was previously determined within the bay in a study carried out during the previous fishing season (Ley-Cooper *et al.* 2013) and is assumed to be constant both over time and the modelled area.

### Analysis of tag data

Data from tagged lobsters were recorded within the MSTR model, using a four-dimensional array that recorded their expected abundance against: (1) their initial release location; (2) the month, they were released; (3) their recapture location; and (4) the month of recapture. At the start of each time-step, tagged lobsters were recruited into the model using the equation:

$$N_{r,m,a,t} = T_{r,m} + N_{r,m,a,t-1} \quad (1)$$

where  $N_{r,m,a,t}$  represents the number of tagged lobsters from the release area ( $r$ ) (area II), released in month  $m$  and currently residing in area  $a$  (area I), during the model time-step  $t$ , and  $T_{r,m}$  represents the number of tagged lobsters initially released:  $m$  and  $t$  range from May to January ( $n = 11$ ), while  $r$  and  $a$  represent lobsters moving from deep to shallow ( $n = 2$ ).

### Migration

Following the recruitment of tagged lobsters, the MSTR model estimated a time-step-specific proportion of tagged lobsters ( $P_t$ ) that were uni-directionally migrating from the unfished area to the fished area using the equations:

$$N_{r,m,1,t} = N_{r,m,2,t} * P_t, \quad (2)$$

$$N_{r,m,2,t} = N_{r,m,2,t} * (1 - P_t). \quad (3)$$

### Estimates of mortality, recoveries and tag loss

Estimates of  $F$  and  $M$  were based on previous estimates determined for this same fishery in the previous fishing season 2010/2011 and were assumed to remain constant across the fishing season (Ley-Cooper *et al.* 2013). These values were not determined by the MSTR model as no tagging occurred within the fished area (I). As the average monthly estimate for  $F$  had been determined for the previous fishing season at  $\sim 0.3$  (Ley-Cooper *et al.* 2013), a range of values that encompassed this (e.g. 0.1, 0.15, 0.2, 0.25, 0.4, and 0.5) were used to determine how sensitive the model was to the estimate of  $F$ . The instantaneous rate of natural mortality ( $M$ ) of  $0.02\text{-month}^{-1}$  used in this study was also taken from the Ley-Cooper *et al.* (2013).

The rate of tag loss used in the model (about  $14\% \text{ month}^{-1}$ ) was based on the rate previously reported for *P. argus* in the same fishery (Ley-Cooper *et al.* 2013), and it represented the expected reduction in the number of tagged lobsters at liberty, due to tag shedding and tag induced mortality. After allowing for the migration of lobsters between areas, 50% of the estimated monthly tag loss was applied to the simulated population of lobsters both before and after fishing, and natural mortality had

both been applied. Tag loss was applied to the simulated population of lobsters in the model by using the equation:

$$N_{r,m,a,t} = N_{r,m,a,t}(1 - e^{-2.65}). \quad (4)$$

Interviews with fishers indicated that the tag reporting rate for this study was  $\sim 30\%$  less than that of the previous seasons estimate of 100% return rate (Ley-Cooper *et al.* 2013). This was considered to be due to less direct interaction with the fishers and that tag reporting was becoming less novel. As such, a range of tag reporting rates encompassing this estimate (60–80%) were trialled in the model to assess its sensitivity to this parameter.

Instantaneous rates of fishing ( $F$ ) and natural mortality ( $M$ ) were applied to the population, once tagged lobsters had been released, migrated and reduced in magnitude through tag loss. Simulated catches were also affected by a tag reporting rate ( $\lambda$ ) which was applied before the lobsters were considered to have been reported to the survey team. The estimated number of recaptured lobsters reported to the survey team and the numbers left in the water were determined using a Baranov catch equation:

$$\hat{R}_{r,m,a,t} = N_{r,m,a,t} * \frac{F_{a,t}}{F_{a,t} + M} * \left(1 - e^{-(F_{a,t} + M)}\right) * \lambda_{a,t}, \quad (5)$$

$$N_{r,m,a,t} = N_{r,m,a,t} * e^{-(F_{a,t} + M)}, \quad (6)$$

where  $\hat{R}_{r,m,1,t}$  is the estimated reported catch of tagged lobsters and  $M = 0.24/12\text{-month}^{-1}$ . Note fishing mortality in the unfished area is assumed to be zero ( $F_{1,t} = 0$ ).

### Likelihood of tag recoveries

A nonlinear function minimisation procedure was used to estimate the values of the six migration parameters that maximised the likelihood of the observed tag recoveries. The parameters estimated represented the month-specific proportion of lobsters migrating from the unfished to the fished area, in each month from August to January ( $P_t = 1-6$ ). Initial parameter estimates used were 0.05 for each of the 6 months, that is, 5%, of tagged lobsters migrated in every month. The negative log-likelihood (LL) of the observed recoveries of tagged lobsters ( $R$ ) given by the monthly proportions of lobsters migrating ( $P_t$ ) and the MSTR model was assumed to have a Poisson distribution (Hilborn 1990) and was represented by

$$LL = - \sum \left( -\hat{R}_{r,m,2,t} + R_{r,m,2,t} \left( \ln(\hat{R}_{r,m,2,t}) \right) - \ln(R_{r,m,2,t}!) \right). \quad (7)$$

The log-likelihood was maximised using the 'optim' routine in 'R', with the square-root of the diagonal of

the inversed hessian matrix, being produced to approximate the standard errors of the parameter estimates.

## Results

### *Size class differences between unfished areas (>20 m) and fished areas (<20 m)*

In Bahía del Espíritu Santo, there was a significant difference ( $P < 0.001$ ) in size composition between the lobsters found in the deeper unfished area (>20 m, area II) and those in the shallower commercially fished bay (area I). In the deeper area, the mean CL of lobsters was 94.2 mm CL, with 99% of lobsters being larger than the minimum legal size (74.5 mm CL). By contrast, lobsters in the fished area (area I) had a mean CL of 73.0 mm, and 75% were smaller than the minimum legal size CL (Fig. 2).

### *Movements*

Of 379 lobsters released, 20 were recaptured (5.3%) within the fishery (area I) during the 2011/2012 fishing season. A further four lobsters were recaptured within the fishery during the subsequent 2012/2013 fishing

season (total 6.3%). Recaptured lobsters comprised 50% females (average size: 85.9 mm CL, size range: 79.1–109.9 mm CL) and 50% males (average size: 101.4 mm CL, size range: 82.2–114.5 mm CL) and were caught over seven separate months starting from October 2011 ( $n = 10$ ), November 2011 ( $n = 8$ ), December 2011 ( $n = 1$ ), January 2012 ( $n = 1$ ) towards the end of the first season and then July 2012 ( $n = 2$ ), November 2012 ( $n = 1$ ) and January 2013 ( $n = 1$ ) towards the end of the second season. Recaptured lobsters had mostly travelled in a southwest direction over distances ranging from 3.5 to 29.2 km when measured in a straight line, with a mean distance of 7602 m (Fig. 3). The lobsters recaptured in October 2011 were located near the fringing reef, whereas those recaptured in November and December 2011 had moved far greater distances and were found closer to the centre of the bay (Fig. 3). All of the lobsters recaptured in the second season following tagging (2012/2013) were captured near the fringing reef on the outside edge of the bay (Fig. 3).

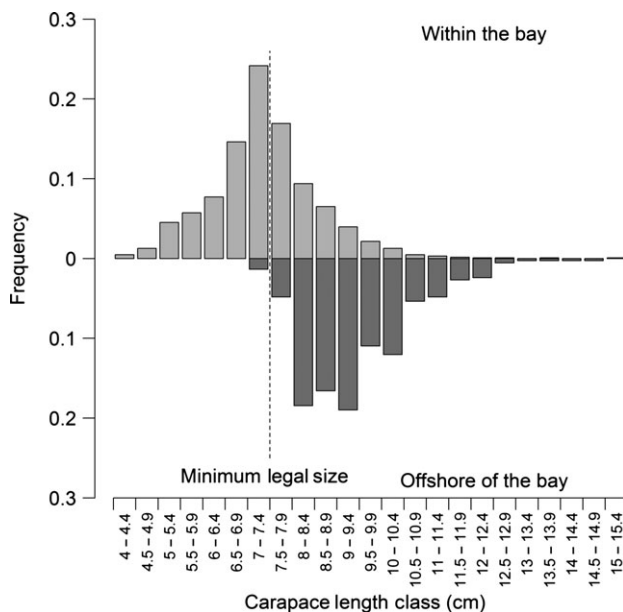
### *Catch trends, effort and catch rates for the inshore fished area*

In Bahía Espíritu Santo, catch trends follow a similar pattern throughout the years, starting at the highest levels at the beginning of the fishing season in July, and progressively declining as the season advances towards the end in February. Catch-per-unit-effort (CPUE) is also highest in July, after which it remains relatively constant from August to November, except for a second peak during the autumn period. This latter peak in CPUE generally occurs in October or November and may be associated with the onset of the Nortes (cold fronts arriving from the north).

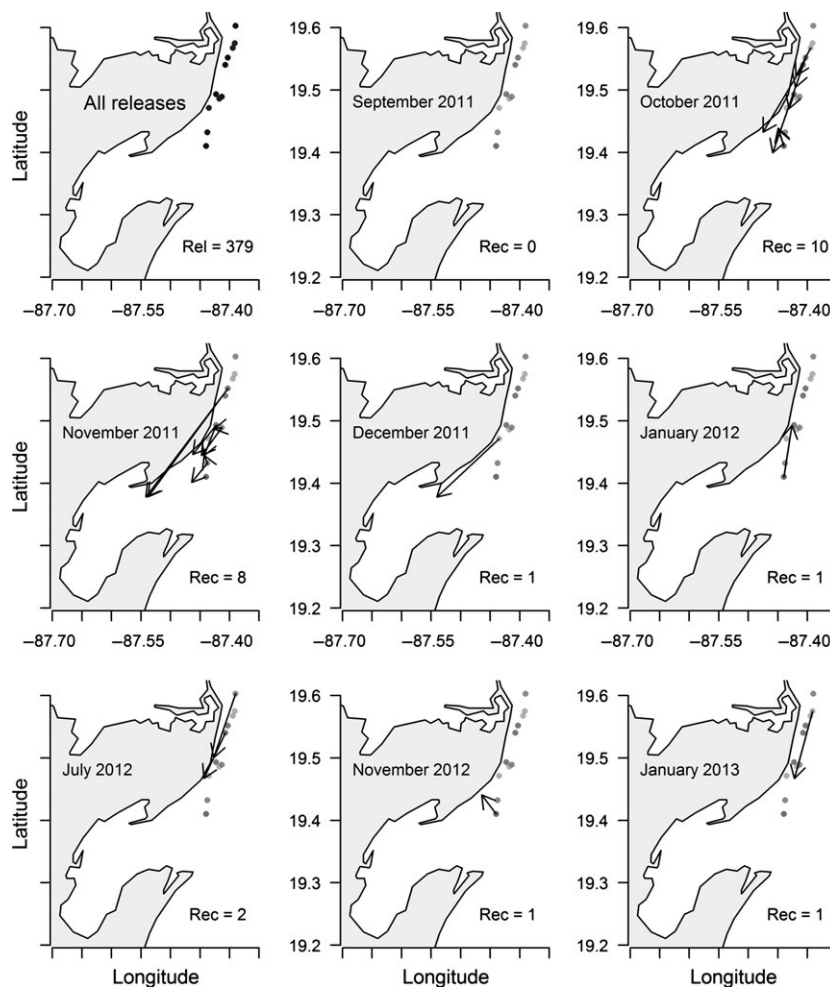
### *Outputs of the model*

Captures of lobsters from the two release pulses (i.e. August and September 2011) displayed similar patterns during the 2011/2012 fishing season, with no recoveries being reported prior to a peak in recaptures in October/November before progressively reducing through until January 2012 towards the end of the first fishing season. The MSTR model was able to recreate a very similar distribution of the tag-recapture pattern observed in the fishery during the 2011/2012 season, also estimating tag recaptures peaked in October 2011 and declined slowly through until January 2012 (Figs 4 & 5).

Under all sensitivity scenarios, a consistent pattern of migration from the unfished area (area II) into the fished area (area I) was estimated to have occurred as a single pulse during the month of October (Fig. 4a). The



**Figure 2.** Size frequency composition of *Panulirus argus* lobsters sampled within Bahía del Espíritu Santo, in the Biosphere Reserve Sian Ka'an- Mexico. Relative frequencies of carapace length (CL) in the fished shallow waters (<20 m) are depicted in light grey (top), and unfished offshore areas (depths >20 m) in darker grey (bottom). The dotted line crosses the X axis showing the boundary between illegal and legal sized lobsters carapace length CL (74.5 mm CL).



**Figure 3.** Bahía del Espíritu Santo (Southern bay) and Bahía de la Ascensión (Northern bay), in the Biosphere Reserve Sian Ka'an Mexico where lobsters were released (dots) and recaptured (arrows point) for each month in which the tagging programme took place. Arrows show the distance and direction of lobsters travelling from the unfished areas (>20 m) where they were tagged, into Bahía del Espíritu Santo.

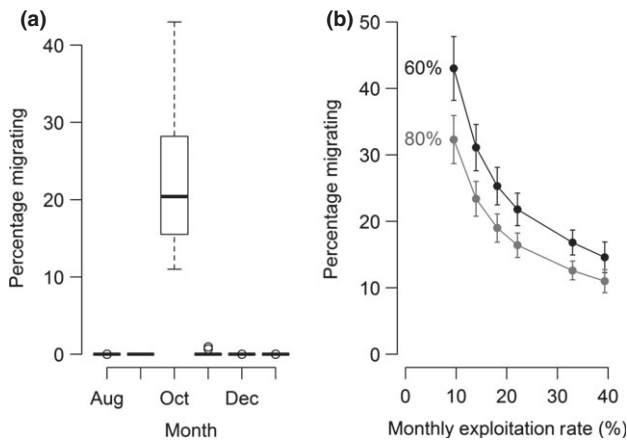
proportion of lobsters estimated to have migrated had a median value of 20% and was relatively consistent across the majority of the scenarios tested for the different variables, that is, ranging from 10 to 40%. The largest percentage of lobsters estimated to be migrating (~40%) occurred under the scenario of a low tag reporting rate (60%) and low exploitation rate (10%) (Fig. 4b).

## Discussion

Results suggest that legal sized lobsters (subadults 74.5–80.0 mm CL and adults >80.0 mm CL) move from the offshore unfished area and enhance the commercially fished area in the shallows within the SK-BR. The movement rates of the tagged lobsters were examined using a purpose built multistate tag-recapture model,

which was able to replicate the observed tag recoveries obtained from the commercial fishery. Based on an assumed monthly rate of fishing mortality 30% (which was reported the previous season, e.g. Ley-Cooper *et al.* 2013) and a rate of tag reporting of 70%, about 20% of the lobsters originally dwelling offshore migrated into the bay. Sensitivity analysis indicated that if the assumptions for fishing mortality and tag reporting were varied within sensible ranges, the proportion estimated to be migrating did not vary radically, generally remaining within the range of 15–20%.

The model estimated that the movements all occurred within the same month (October 2011), a period within that year that coincided with the start of a cold front system, as well as an annual increase in the catch-per-unit-effort in the commercial fishery of both the northern and southern bay of the Biosphere Reserve (K. Ley-Cooper,

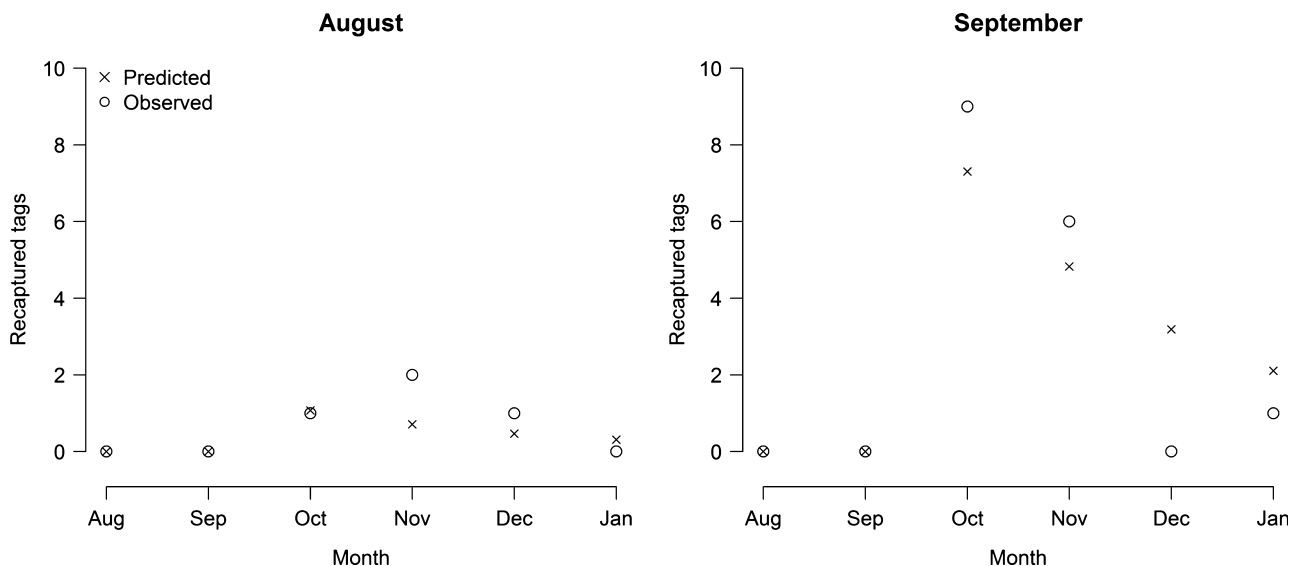


**Figure 4.** Multistate tag-recapture model modified from Hilborn (1990) and sensitivity analysis for the most likely exploitation rates and tag reporting scenarios. (a) Left panel: most likely percentage of lobsters that migrate from the unfished areas II towards the fished shallow bay (area I), with a mean value of 20%, and range between 40–10%. Scenarios are based on data which replicate the observed tag recoveries under the most likely exploitation rates from the sensitivity analysis. The model simulations show that October is when most lobsters moved in one pulse, with a small probability of movements in the other months (outliers are less than 2%). (b) Right panel: Percentage of lobsters which migrated from the deep unfished areas II, towards the fished shallow bay (area I). The span of range values used in the sensitivity analysis was based on the most likely estimates for exploitation rates (0–40%) and tag reporting rates (60 and 80%) resulting in the simulation scenarios on the black and grey lines.

unpublished data). The movement of spiny lobsters has been well documented for many spiny lobster species and has been generally categorised into three types: homing,

nomadic or migratory movements (Herrnkind 1980; Wahle & Fogarty 2006). Migratory movements are unidirectional, occur in mass quantity, during a confined period of time, and have been attributed to several factors such as ontogenetic behaviour; seasonal movements towards feeding grounds (Acosta 1999; Briones-Fourzán *et al.* 2003; Ríos-Lara *et al.* 2007); spawning migrations (Bertelsen 2013); or as a response to environmental stimuli such as moon cycles, changes in water temperatures and changes in wind strengths and direction (García *et al.* 1991). In Florida, Bahamas and Cuba, movements of *P. argus* were reported to be initiated by cold fronts in autumn, as increasing wind speed and direction increased water turbidity and decreased water temperature, which in turn triggered lobster mass migration behaviours (Herrnkind 1980, 1985; García *et al.* 1991). Data from 1984 to 2012 from the northern bay of SK-BR-Bahía de la Ascensión showed that an increased meridional wind speed and change of direction results in a direct relationship with increased catch rates during the autumn period (K. Ley-Cooper, unpublished data). Using CPUE as an indicator of abundance, a plausible hypothesis is that cold fronts generate additional nomadic movements of lobsters due to changes in temperature and increased turbidity which allows for greater foraging distances, which may result in movements from the deep (area II) to the shallow bay (area I). The model estimated that most movements occurred as a single pulse, which suggests that a migration-like behaviour could have been the causative factor.

Regardless of the motive, the lobster movements (mainly large adults see Fig. 2) from an unfished off-



**Figure 5.** Multistate tag-recapture model outputs showing distribution of the tag-recapture patterns observed in the fishery. Data were obtained from lobsters tagged in unfished areas II offshore and recovered in the fished shallow bay (area I) in the study site of Bahía Del Espíritu Santo Sian Ka'an. The panels show the fitting of the observed against the predicted tag-recapture patterns. The left panel corresponds to the batch of lobsters released during August, and the right panel to the batch released in September within the unfished area II offshore.

shore area into the shallower commercially fished bay are contrary to the general pattern of migration observed in spiny lobsters, as the paradigm is usually small immature lobsters moving offshore as part of their normal ontogenetic behaviour (Briones-Fourzán *et al.* 2003; Melville-Smith & de Lestang 2006; Wahle & Fogarty 2006). Movements of juveniles and adults within the bay and towards offshore areas have also been documented in previous tagging studies in this area (Ley-Cooper *et al.* 2013), yet the migration of these large lobsters from the deep offshore towards the shallow bay had not previously been documented.

In a previous study conducted in Bahía de la Ascensión (BA) (see Fig. 1), Lozano-Álvarez *et al.* (1993) suggested that adult lobsters probably returned to that shallow bay after breeding in offshore areas of the deeper shelf. González-Cano (1991) analysed recruitment using size structure and catch data, suggesting that seasonal migrations from deep to shallow areas could occur annually as lobsters are redistributed in Isla Mujeres (further north) (Fig. 1). The latter study supports the findings in this assessment, but the use of a multistate tag-recapture model as presented here had additional benefits, as it was able to provide an estimate of the proportion of the population moving into the bay, a measure that is particularly useful for conducting spatial stock assessments (Ziegler *et al.* 2003; Goñi *et al.* 2010; McGarvey *et al.* 2010).

Lobsters previously tagged in Bahía de la Ascensión (BA) were also recaptured in Bahía del Espíritu Santo. These movements took 2 years, with lobsters covering a distance of about 43.5 km (Ley-Cooper *et al.* 2013). Whether the lobsters tagged for this study originated in BA when they started to migrate remains unknown, yet the estimated 15–20% proportion indicates that recruitment into these fisheries from deep waters occurs and that there is some level of offshore adult connectivity after 20 m depths between both bays of the SK-BR.

This migration also provides new evidence for the potential maintenance or enhancement of fishing yields into areas which are adjacent to unfished reserves. In the two bays of the SK-BR, maximum catch rates are observed as peaks at the beginning of the fishing season in July and in the mid-season during autumn, between October and December (Sosa-Cordero *et al.* 2008; Ley-Cooper *et al.* 2013). Fishing exploitation rates are high within the shallow-bay area, and results of this study suggest that to sustain such high catch rates in autumn, lobsters recruited to these fisheries must be partially replenished by both juvenile growth (moulting from undersize to legal sized lobsters) (González-Cano 1991; Lozano-Álvarez *et al.* 1991b, 1993; Briones-Fourzán *et al.* 2007; Ley-Cooper *et al.* 2013) and by the input of lobsters moving in from unfished areas found offshore in

deeper waters (>20 m). The contribution to catch rates within Bahía del Espíritu Santo that were derived from growth, movements, natural and fishing mortality had been previously explored and reported (Sosa-Cordero *et al.* 1999; Ley-Cooper *et al.* 2013), yet the proportion of migrating lobsters from the deeper unfished areas had not been estimated or described as a source of recruitment.

The significant difference in the size composition of lobsters between the deep unfished and shallow fished areas may be attributed either to ontogenic movements such as juvenile progressive growth and migration offshore (Lozano-Álvarez *et al.* 1993) and/or the high rates of fishing mortality of legal sized lobsters which only occurs within the commercially fished bay, as causes of natural mortality are generally independent of depth and most likely similar in both areas (Ley-Cooper *et al.* 2013).

A number of factors could have biased our estimates of migration in this study, including the relatively small sample size of recaptured lobsters. It would be advantageous, should the study be repeated, if a significantly greater number of lobsters were to be tagged. Tag loss is also an area where a bias may be incorporated. The rate of tag loss used in this study was based on that determined from aquaria trials and therefore may not directly mimic that which may have occurred in the reserve during this study. Different water quality, lobster densities and habitat availability all could have caused the tag loss rate to have differed. A decreased rate of tag loss would have biased the results and increased the estimate of the proportion of lobsters migrating. Future work in this area could include the double tagging a number of the individuals released to examine *in situ* tag loss.

Although tags continued to be returned during the following season (2012/2013), their numbers were too low to be added into the model. The two tags returned at the start of the second season may have been from lobsters that migrated into the bay either during the closed season (February–June) or during the previous October (2011) and survived fishing mortality during the remainder of that season. It is interesting that no other tagged lobsters were recaptured within the bay until November 2012, 5 months into this second season, at a point when over 70% of the season's annual catches had been landed. It is possible that the lobsters recaptured in November 2012 and January 2013 did not enter the bay during October 2011, but rather during a second autumn migration occurring in the October/November 2012. The lack of robust information provided by these 2012/2013 tag returns highlights the value of continuous studies of this nature. A series of yearly multi-release tagging campaigns in both the fished and unfished areas are

recommended for the future, which could provide annual estimates of the overall biomass contributed to the fishery by the lobsters migrating into the bay.

According to the model produced in this study, the larger proportion of the deep segment of the population remains unfished, and it is suggested that in the SK-BR, the deep areas (>20 m) should remain as such. To guarantee a sustainable management of this fishery and the conservation of the *P. argus* lobster population, it would be advantageous to understand the variations on the yearly proportions of large lobsters moving from the unfished areas into the shallow areas subject to fishing and the effects that this may have on the biomass of the total stock. It is recommended that these issues be further addressed in future studies.

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